Content

Learning objectives and content

Objectives

1. Students will be able to compare major software development processes and to argue about which process to apply in different scenarios.
2. Students will be able to practice the use of version control systems (VCSs) by developing a software solutions individually and as a small team.
3. Students will be able to translate requirements into specifications that peers can understand and use.
4. Students will be able to employ software design principles & design patterns while developing a software solution and to argue when these are feasible.

Content
Students of other programs can be enrolled by the academic advisors upon approval y the responsible lecturer (j.kruger@tue.nl)

We assume basic knowledge of Java—as treated in the course 2IP90 Programming, or at a similar level, covering the basic concepts of imperative programming, objects, classes, and inheritance. We also assume basic knowledge about GUI programming using Java Swing. If you do not have this particular background, you should self-study Chapter 5 Objects and Classes and Chapter 6 Introduction to GUI Programming of the free e-book “Introduction to Programming Using Java” by David J. Eck (Version 9.0, Swing Edition, May 2022). This book can also serve as a refresher and way of extending your existing Java knowledge.

If you have a background in, for instance, Python or C++, the basic concepts mentioned should already be familiar to you. For the specifics of Java, the first four chapters of David Eck's book can serve as an introduction and reference. If you know Python, you can also use the shorter “java4python” by Brad Miller and James Riely (Release 3.0, February 24, 2015). With some self-studying, your lack of Java background should not be a problem for following and passing the course.

2026

Software Design (2IRR00)

Practical information

Course module
2IRR00

Course type
Bachelor College

Credits (ECTS)
5 EC

Category
Deepening

Language of instruction
English

Offered by
Mathematics and Computer Science - Computer Science domain

Remark
Enrolment restrictions apply for non-BCS students and BCS students cohort 2022 and earlier. Please check content

section

Short promotional description of the course
Defining a feasible software design is important to prevent bugs, rework, and maintenance problems. While important, it is also highly challenging to define a proper software design. Many design processes, paradigms, and patterns exist, each having different pros and cons. In this course, we will provide an overview of software-development processes, version-control, requirements engineering, design principles, graphical user interfaces, and design patterns focusing on object-oriented software development with Java. Involving lectures, exercises, and graded individual as well as group work assignments, the course aims at providing the theoretical foundations needed to properly design software, and at enabling students to explore these on smaller example projects.

Course enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 3
  Timeslot(s)
  A - Mo 1-4, We 9-10, Th 5-8

  Enrolment period
  15 November 2026 until 3 January 2027 23:59

  Unenrolment
  until 1 February 2027 23:59

Entrance requirements

Assumed previous knowledge
It is assumed that students participating in this course have completed a course on object-oriented programming (such as 2IP90). Only BCS students can register for this course directly. Other students who want to enrol should contact the responsible lecturer (j.kruger@tue.nl), specifying how they satisfy the requirement of having completed a course on object-oriented programming.

Entrance requirements course enrolment
You must meet the following requirements

- Enrolled for one of the following degree programmes
  - Computer Science and Engineering
- Completed none of the course modules listed below
  - Programming Methods (2IPC0)

Instructional modes

Instructional modes

- Instruction
  Remark
- Lecture
  Remark

Tests

Tests

- Assignment
  Interim examination

  Test weight
  35

  Minimum grade

- Schoolyear ANS
  Final examination

  Test weight
  55

  Minimum grade
  5.0

- Written paper based ANS test
  Interim examination

  Test weight
  10

  Minimum grade

Test enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignment

- Block 3
  Enrolment period
  15 November 2026 until 21 March 2027 23:59

Schoolyear ANS

- Block 3
  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 21 March 2027

  Test date
  5 April 2027

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

  Unenrolment
  until 6 June 2027

  Test date
  28 June 2027

Written paper based ANS test

- Block 3
  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 21 March 2027

Lecturer(s)

Contactperson for the course

- J. Krüger

Responsible lecturer

- J. Krüger

Subject matter expert

- J. Krüger

Materials

Recommended material

- Lecture slides as script
- Martin P. Robillard: “Introduction to Software Design with Java”
- Brahma Dathan and Sarnath Ramnath: “Object-Oriented Analysis, Design and Implementation: An Integrated Approach”
- Martina Seidl, Marion Scholz, Christian Huemer, and Gerti Kappel: “UML@Classroom: An Introduction to Object-Oriented Modeling”

Do you want to know more?

J. Krüger can tell you more about this course.
Send an e-mail to j.kruger@tue.nl